Cerrado Tree Species Identification for Aerial Robots: A Dataset and Embedded YOLO Evaluation

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***Abstract*—Tree species identification from aerial images of tree crowns can support forest inventories, biodiversity monitoring, ecological restoration, and environmental management. However, deploying such capabilities on aerial robotic platforms requires supervised models trained on reliable datasets, whose construction involves fieldwork, taxonomic expertise, and substantial annotation effort. This study presents an integrated workflow for identifying Cerrado tree species from UAV RGB images, combining field-validated taxonomic data, assisted annotation, instance segmentation, and edge-computing evaluation. The Segment Anything Model 3 (SAM3) was used to assist tree-crown annotation and support the construction of a curated dataset of Cerrado tree species. You Only Look Once (YOLO) models were trained and evaluated for tree-crown instance segmentation and species identification. Different data preparation strategies, including the use of negative images, were investigated to assess their influence on model performance. The trained models were also evaluated on an embedded computing platform to assess inference performance and the potential impact of onboard processing on UAV mapping operations. The results indicate that the proposed workflow can support the construction of datasets and the development of computer vision models for tree species identification, while the embedded evaluation provides an initial assessment toward future deployment on aerial robotic platforms for forest inventory and environmental monitoring.**

I. Introduction

In forest inventories and environmental monitoring, vegetation characterization involves identifying species and determining their spatial distributions, rather than merely estimating cover. This information supports ecological research, conservation, sustainable management, the protection of vulnerable populations, invasive species control [1]–[3], ecological restoration [4], and the mapping of socioeconomically valuable resources [5]. However, traditional field-based surveys require substantial effort and financial resources while providing limited spatial coverage, thereby hindering large-scale and continuous monitoring [6].

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UAV-based remote sensing combined with computer vision offers a promising approach to expanding vegetation monitoring capabilities. UAVs provide high-spatial-resolution imagery and greater operational flexibility than crewed aircraft, while offering finer spatial detail than satellite-based observations [7]. These capabilities enable the analysis of individual tree crowns and support the automation of tree localization, segmentation, and classification, potentially improving monitoring efficiency and reducing reliance on exclusively ground-based sampling.

In the Cerrado, these technological advances are particularly relevant because the biome covers 23.9% of Brazil [8] and is recognized as a global biodiversity *hotspot* due to its high species richness, endemism, and phytophysiognomic heterogeneity [9], [10]. Despite its ecological significance, the Cerrado accounted for 54.9% of the native vegetation suppression recorded in Brazil in 2025 [8]. Within this biome, automated tree species identification remains challenging because of crown morphological variability, overlapping individuals, and spectral and visual similarities among species [1].

Although deep learning offers promising approaches to this task, its effectiveness depends on the availability of reliable, carefully annotated datasets. Constructing datasets for tree species identification requires linking field-identified individuals, their geographic coordinates, and their taxonomic records to their corresponding crowns in UAV imagery. This process involves expert validation and manual crown delineation, making annotation labor-intensive and limiting the scalability of automated mapping systems.

To address this bottleneck, this study proposes an assisted-annotation workflow that integrates field-based taxonomic records and UAV RGB imagery, employing the *Segment Anything Model 3* (SAM3) [11] to support tree-crown segmentation. SAM3 generates preliminary individual-crown masks from prompts, reducing manual delineation effort and supporting the association of image segments with taxonomic labels. Using this workflow, a curated dataset of Cerrado tree species was constructed from UAV RGB imagery and field-validated crown annotations.

To assess the dataset's suitability for supervised tree species identification, models from the *You Only Look Once* (YOLO)

family were trained and evaluated on the annotated images. These models integrate object localization, instance segmentation, and classification within a unified architecture. The experiments investigate the effects of tile size, negative-image inclusion, and data augmentation on model performance. In addition, inference was evaluated on an edge-computing platform to assess the feasibility of deploying the proposed approach in future aerial robotic applications for environmental monitoring.

Thus, the main contributions of this study are threefold: (i) a SAM3-assisted workflow for constructing annotated tree-crown datasets from UAV RGB imagery and field-validated taxonomic records; (ii) the construction and evaluation of a Cerrado tree species dataset using YOLO-based instance segmentation, considering tile size, negative-image inclusion, and per-species performance; and (iii) an assessment of edge-computing inference performance to determine the feasibility of deploying the proposed approach on aerial robot platforms.

II. Dataset Construction Workflow

This section describes the SAM3-assisted workflow developed to construct annotated datasets from UAV RGB imagery. The pipeline integrates field-validated taxonomic records with aerial images and employs foundation model-assisted segmentation to generate annotations suitable for computer vision tasks.

As shown in Figure 1, the pipeline comprises a sequence of stages. First, RGB images are acquired using UAVs (1), and field records of the target trees are collected (2) to establish the taxonomic reference data. The images are then filtered to retain those containing field-recorded specimens (3), and the corresponding georeferenced points are overlaid on the selected images (4). Based on this spatial association, SAM3 assists in generating initial segmentation masks (5). These masks can be manually refined when necessary (6) and assigned the corresponding species labels using the field records (7). Finally, the annotated dataset is prepared for model training by incorporating negative samples (8), dividing the images into overlapping tiles and creating training, validation and test subsets (9), and exporting the annotations in formats compatible with instance segmentation models (10). The application of this workflow to the Cerrado tree species dataset is detailed in Section III.

A. Data Acquisition, Field Reference Data, and Image–Field Integration

The pipeline combines UAV RGB imagery with georeferenced field-validated taxonomic records. Spatial intersection filters acquired images to retain only frames covering surveyed specimens, whose coordinates are then projected onto the image plane.

B. SAM3-Assisted Annotation

Following image–field integration, SAM3 is used to generate candidate segmentation masks from textual or visual prompts. The resulting masks are inspected by an operator and may be refined, discarded, or supplemented when the segmentation is unsuccessful or the resulting boundaries are inaccurate. For crowns that are not adequately segmented, or for geometrically complex cases, a bounding box can be defined around the region of interest to guide segmentation toward the target crown. The validated masks are subsequently associated with their corresponding taxonomic attributes from the field records, forming the initial set of ground-truth annotations.

C. Dataset Generation

After annotation, the base dataset is prepared for model training. Negative images, which contain no instance of the target species, can be incorporated to increase background diversity and help reduce false-positive detection in complex environments. The high-resolution images are then sliced into overlapping square tiles. The resulting tiles are assigned to training, validation, and test subsets at the image level to prevent data leakage between them. Finally, the images and annotations are converted into a format compatible with YOLO-based instance segmentation models.

III. Cerrado Dataset Construction

Following the workflow described in Section II, this section details its application to the construction of a Cerrado tree species dataset. The description covers image acquisition, field validation, annotation, class selection, and dataset preparation for species-identification experiments.

A. Image Acquisition and Field Validation

The images were acquired in a Cerrado area in Brazil using a DJI Matrice 350 RTK UAV equipped with a DJI Zenmuse L1 sensor. The flight was conducted at an altitude of approximately 60 m above ground level, with a speed of 7 m/s and 60% forward and lateral image overlap. Although the sensor supports both LiDAR and RGB acquisition, only the georeferenced RGB images were used in this study. The RGB camera produces 5472×3648-pixel images, yielding an estimated ground footprint of 89.95×59.97 m before accounting for image overlap and a theoretical ground sampling distance (GSD) of 1.64 cm/pixel. In the final dataset used for annotation, the average observed GSD was 2.5 cm/pixel.

Field reference data were collected by taxonomic specialists, including a biologist and a forest engineer. Each tree was identified, georeferenced with a base–rover real-time kinematic (RTK) setup, photographed, and recorded in an Android application. The point identifiers and corresponding species labels were exported as a GeoPackage vector layer (`.gpkg`), providing the spatial reference data for associating field observations with the aerial imagery.

B. Annotation and Base Dataset

The annotation process began with spatial filtering based on the intersection between the acquired images and the GeoPackage layer. This step retained only image frames containing field-validated trees, excluding aerial coverage without

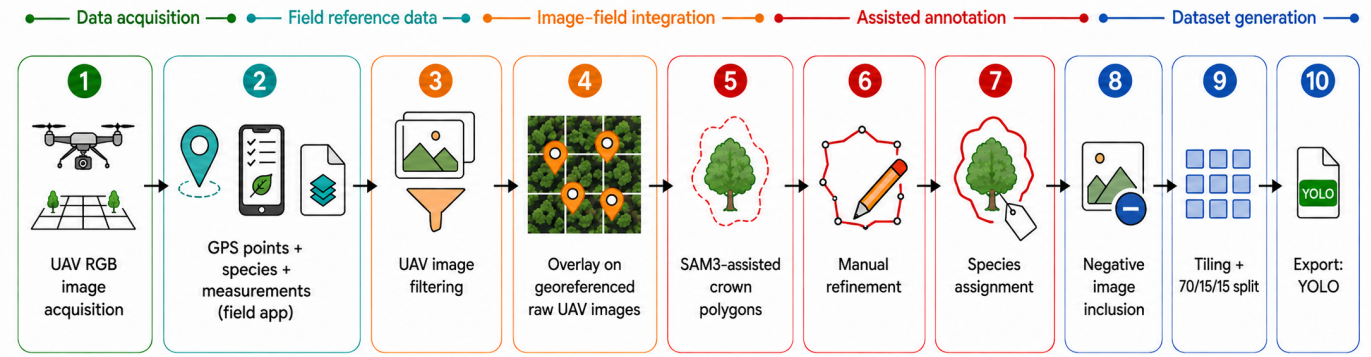


Fig. 1. Overview of the proposed workflow, from UAV image acquisition and field reference data to SAM3-assisted annotation and YOLO-compatible dataset generation.

corresponding field references. The associated vector points were subsequently overlaid on the selected images to help locate individual tree crowns.

A custom annotation interface integrating SAM3 was used to annotate the Cerrado tree species (Figure 2). Initial candidate masks were generated using the textual prompt *tree*. The operator reviewed the resulting masks and discarded false positives that could not be associated with field records. For crowns that were not successfully segmented or that exhibited complex shapes, bounding boxes were used to define regions of interest and to guide mask generation, thereby avoiding the need for manual vertex-by-vertex delineation. Each approved polygon was assigned the corresponding taxonomic label, resulting in the base annotated dataset.

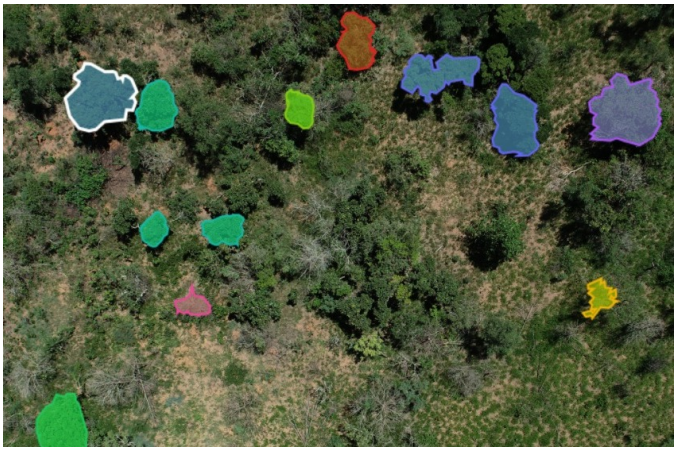


Fig. 2. Annotation interface used to support SAM3-assisted mask generation and curation.

C. Class Distribution and Curation

To mitigate class imbalance and limited representation [12], species with few instances were excluded, resulting in a curated dataset of 136 instances across 7 classes. An instance denotes an annotated crown occurrence per image, meaning individual trees appearing in multiple UAV frames contribute multiple instances. Table I details the species distribution.

TABLE I
DISTRIBUTION OF ANNOTATED INSTANCES BY SPECIES IN THE CURATED CERRADO DATASET USED IN THE EXPERIMENTS.

Species	Inst.	%
Diospyros lasiocalyx	53	38.97
Annona crassiflora	22	16.18
Handroanthus ochraceus	16	11.76
Vochysia tucanorum	16	11.76
Eriotheca gracilipes	13	9.56
Leptolobium elegans	9	6.62
Dimorphandra mollis	7	5.15
Total	**136**	**100.00**

D. Data Preparation for Training

To train YOLO instance segmentation models, overlapping square tiles were generated from high-resolution aerial images at resolutions of 192, 256, 384, 512, 768, 1024, and 1280 pixels. The dataset was split 70/15/15 into training, validation and test sets at the original-image level to prevent data leakage between tiles derived from the same image.

Two dataset variants were evaluated: Cerrado-V1, comprising original images containing field-validated trees, from which background-only tiles could naturally arise during tiling; and Cerrado-V1-Neg, which additionally included negative samples containing no target species. These negative images were acquired at a different Cerrado site with a similar GSD (~2.5 cm/pixel) and included exposed soil, low vegetation, shadows, and non-target trees. Their inclusion was intended to increase background diversity and reduce false-positive detections in complex scenes.

Finally, the vector annotations were converted into the normalized polygon format required by YOLO instance segmentation models.

IV. Experimental Setup

This section describes the experimental protocol used to evaluate the constructed Cerrado dataset for tree species identification using YOLO-based instance segmentation. The experiments investigate model configuration, training parameters, the effect of negative-image inclusion, evaluation metrics, and inference performance on an embedded edge-computing platform.

A. Model Architecture and Training Protocol

YOLOv11-seg models were selected for single-stage instance segmentation, object localization, and species classification, initialized with `yolo11s-seg.pt` pretrained weights. Models were trained across dataset variants using the consistent 70/15/15 image-level split. Training incorporated standard stochastic data augmentation (scaling, rotation, flipping, HSV perturbation, Mosaic, MixUp, and Copy-Paste), with checkpoints selected based on validation performance.

B. Negative Sample Evaluation

Complex backgrounds, shadows, overlapping crowns, and non-target vegetation can increase false-positive detections in UAV imagery [13], [14]. Hard negative examples may help models distinguish target objects from visually similar background structures [15]. To evaluate this effect, Cerrado-V1 was compared with Cerrado-V1-Neg under identical model architecture, training parameters, data split, and number of epochs.

C. Evaluation Metrics

YOLOv11-seg predictions were evaluated against reference masks using precision (P), recall (R), F1-score, mAP@0.5, and mAP@0.5 : 0.95:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F1 = 2\frac{P \cdot R}{P + R}, \quad (1)$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively. Spatial agreement between predicted and reference masks was measured using Intersection over Union (IoU):

$$\mathrm{IoU}(M_p, M_g) = \frac{|M_p \cap M_g|}{|M_p \cup M_g|},$$

in which M_p and M_g denote the predicted and reference segmentation masks, respectively.

Mean Average Precision (mAP) averages the Area Under the Precision-Recall Curve across all C target classes:

$$mAP = \frac{1}{C}\sum_{c=1}^{C} AP_c. \quad (2)$$

We report mAP@0.5 and mAP@0.5 : 0.95, the latter averaged over IoU thresholds ranging from 0.5 to 0.95 in increments of 0.05. Absolute and normalized confusion matrices were also used to assess inter-species misclassifications and background-related false positives.

D. Embedded Inference Setup

To assess embedded deployment feasibility, inference was benchmarked on an NVIDIA Jetson AGX Orin platform with 64 GB of memory. Inference time, total processing time, memory usage, GPU power consumption, temperature, and throughput were recorded. The average total power consumption during mapping was estimated by combining the equivalent UAV–payload power, the continuous Jetson power consumption, and the GPU power contribution during image processing:

$$\overline{P}_{\mathrm{total}} = P_{\mathrm{UAV+L1}} + P_{\mathrm{Jetson}} + P_{\mathrm{GPU}}\left(\frac{t_{\mathrm{pipe}}}{t_{\mathrm{photo}}}\right), \quad (3)$$

where $P_{\mathrm{UAV+L1}}$ denotes the equivalent average power consumption of the DJI Matrice 350 RTK equipped with the Zenmuse L1, P_{Jetson} is the continuous power consumption of the Jetson platform, P_{GPU} is the average GPU power consumption during processing, t_{pipe} is the total processing time per image, and t_{photo} is the image acquisition interval during mapping.

The theoretical flight time was then estimated as follows:

$$t_{\mathrm{flight}} = \frac{E_{\mathrm{bat}}}{\overline{P}_{\mathrm{total}}}, \quad (4)$$

where E_{bat} denotes the nominal energy capacity of the two TB65 batteries.

V. Results and Discussion

This section presents and discusses the YOLOv11-seg results for Cerrado tree species identification, covering tile size, negative samples, per-species performance, qualitative results, and embedded inference.

A. Model Performance and Tile Size

The effect of tile size was evaluated at resolutions ranging from 192 to 1280 pixels for both Cerrado-V1 and Cerrado-V1-Neg. Table II reports precision, recall, F1-score, mAP@0.5, and mAP@0.5 : 0.95 for each configuration.

TABLE II
Performance evaluation of YOLO11s-seg across tile sizes on the Cerrado-V1 benchmark comparing baseline and negative-sample variants (blind test partition, 7 classes).

Tile	Precision		Recall		F1		$mAP_{0.5}$		$mAP_{0.5:0.95}$	
(px)	V1	Neg	V1	Neg	V1	Neg	V1	Neg	V1	Neg
192	0.097	0.962	0.272	0.087	0.143	0.160	0.162	0.133	0.124	0.101
256	0.308	0.138	0.238	0.217	0.269	0.169	0.182	0.148	0.141	0.118
384	0.259	0.514	0.256	0.224	0.258	0.312	0.200	0.250	0.135	0.168
512	0.666	0.835	0.278	0.234	0.392	0.365	0.328	0.252	0.212	0.150
768	0.766	0.397	0.214	0.248	0.335	0.305	0.316	0.268	0.169	0.139
1024	0.719	0.868	**0.377**	0.171	**0.494**	0.286	0.332	0.136	0.211	0.107
1280	0.962	**0.984**	0.317	0.286	0.477	0.443	0.359	**0.372**	**0.218**	0.204

For Cerrado-V1, the highest mAP@0.5 : 0.95 was obtained with 1280-pixel tiles (0.218), followed by 512 and 1024-pixel tiles (0.212 and 0.211, respectively). For Cerrado-V1-Neg, the highest mAP@0.5 : 0.95 was also obtained at 1280 pixels

(0.204), followed by 384 pixels (0.168). These results indicate that larger tiles generally provided a more favorable spatial context for crown segmentation, although performance was not monotonic with tile size.

B. Impact of Negative Samples

The effect of including negative samples was evaluated by comparing Cerrado-V1 and Cerrado-V1-Neg across all tile resolutions using the same test set.

At 1280 pixels, negative samples increased precision from 0.962 to 0.984 and mAP@0.5 from 0.359 to 0.372, while recall decreased from 0.317 to 0.286 and mAP@0.5 : 0.95 from 0.218 to 0.204. These results reveal a precision–recall trade-off, suggesting improved background rejection at the cost of instance recovery. Given its higher precision and mAP@0.5, subsequent analyses focus on the *Cerrado-V1-Neg* model trained with 1280-pixel tiles.

C. Per-Class and Qualitative Analysis

The per-species analysis used the Cerrado-V1-Neg configuration with 1280-pixel tiles, which achieved the highest test precision and mAP@0.5 (0.372). Figures 3 and 4 show its class-wise AP@0.5 and normalized confusion matrix on the validation partition, where performance did not follow class representation.

The confusion matrix shows interspecies confusion and reference crowns assigned to the background (false negatives), mainly for *V. tucanorum* and *L. elegans*.

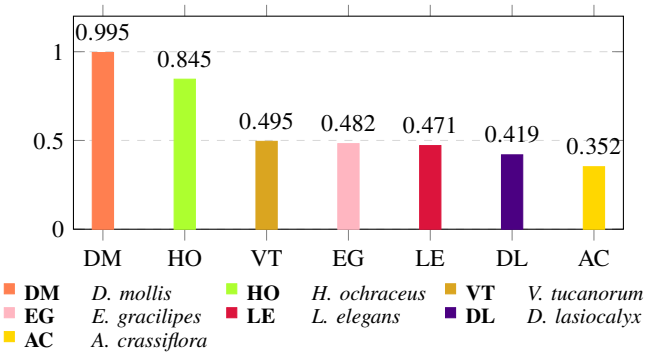


Fig. 3. Class-wise AP@0.5 for the best YOLOv11-seg configuration, trained with Cerrado-V1-Neg and 1280-pixel tiles (validation partition).

Figure 5 presents examples of masks generated by the YOLOv11-seg model in UAV images. These examples complement the quantitative analysis by illustrating the model's behavior in visually complex scenes, including correctly segmented crowns, partial delineations, missed detections, and possible species confusion.

Differences in per-species performance may reflect variation in crown appearance, occlusion, background complexity, and sample availability. Their individual contributions were not quantified in this study.

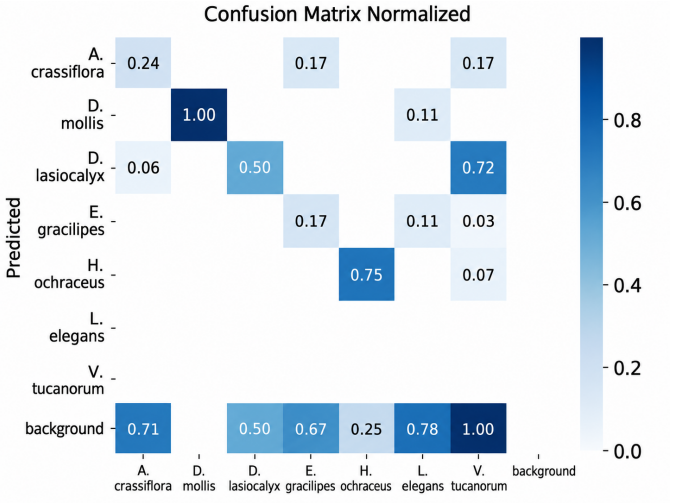


Fig. 4. Normalized confusion matrix of the best YOLOv11-seg configuration, trained with Cerrado-V1-Neg and 1280-pixel tiles.

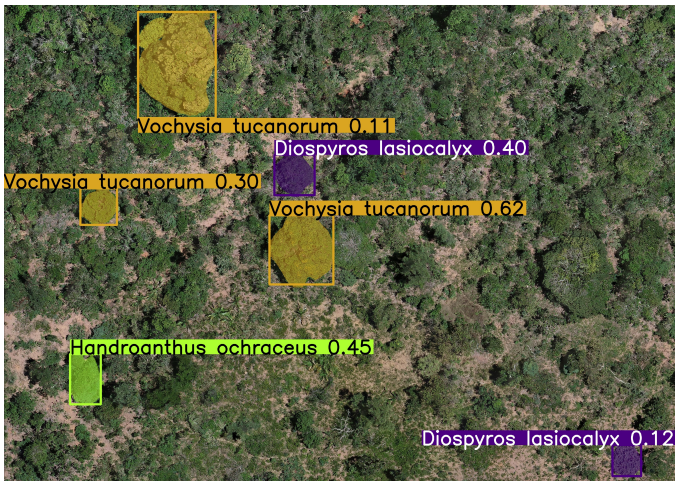


Fig. 5. Qualitative examples of crown masks generated by the YOLOv11-seg model for Cerrado tree species identification in UAV images.

D. Embedded Inference, Flight Endurance, and Coverage Estimation

Embedded inference was evaluated on an NVIDIA Jetson AGX Orin with 64 GB of memory using a 1024-pixel input resolution, selected to reduce computational load relative to the 1280-pixel configuration used in the preceding accuracy analysis. Average inference and complete pipeline times were 0.574 s and 1.266 s per image, respectively. Average memory usage was 12.81 GB, and GPU power consumption was 5.88 W.

Using the nominal energy of two TB65 batteries (526.4 Wh) [16] and a 42-minute baseline flight time, the equivalent M350+L1 power was estimated at 752 W. Including the Jetson load resulted in an estimated total power of 814.17 W and a theoretical flight time of 38.79 min. The corresponding mapping-capacity estimates are summarized in Table III.

TABLE III
ESTIMATED MAPPING CAPACITY WITHIN ONE TB65 BATTERY PAIR.

Configuration	$\overline{P}_{\text{total}}$	t_{flight}	Images	Area
M350 + L1 (Baseline)	752 W	42.00 min	735	63.4 ha
M350 + L1 + Jetson (GPU)	814.17 W	38.79 min	679	58.6 ha

The complete processing time remained below the estimated 3.427 s image acquisition interval. Thus, the main deployment constraints are the continuous computing power demand and added payload mass, which require validation under real flight conditions.

The results should be interpreted considering the limited size and class imbalance of the current dataset. The curated dataset contains 136 annotated crown instances across seven species, with substantial differences in class representation. Moreover, because individual trees may appear in multiple overlapping UAV frames, image-level dataset partitioning may retain correlated observations across subsets. Therefore, the reported results should be regarded as an initial evaluation of the proposed workflow rather than evidence of broad taxonomic or spatial generalization. Future work will expand the number and diversity of field-validated individuals and investigate individual-tree-level or spatially disjoint partitioning protocols for a stricter assessment of model generalization.

VI. CONCLUSION

The proposed SAM3-assisted annotation workflow supported the construction of a field-validated tree-crown dataset from UAV RGB imagery while reducing the need for manual vertex-by-vertex crown delineation. The resulting Cerrado dataset enabled the evaluation of YOLOv11-seg models for tree species identification and instance segmentation under different tile resolutions and background compositions.

The experiments showed that tile resolution and background composition substantially influence model behavior. At 1280 pixels, the inclusion of negative samples increased precision from 0.962 to 0.984 and mAP@0.5 from 0.359 to 0.372. However, recall decreased from 0.317 to 0.286 and mAP@0.5:0.95 decreased from 0.218 to 0.204. These results indicate that negative samples can improve the model's ability to reject background regions, but also reveal a precision–recall trade-off that should be considered when constructing datasets for tree species identification. Per-class and error analyses further showed considerable variation among species, with false negatives and interspecies confusion remain a relevant challenge.

Finally, the embedded evaluation on the NVIDIA Jetson AGX Orin showed that the segmentation pipeline can operate within the image acquisition interval considered in the UAV mapping scenario. The complete processing pipeline required an average of 1.266 s per image, compared with an estimated 3.427 s image acquisition interval. Under the adopted power assumptions, embedded processing reduced the theoretical flight time from 42.00 to 38.79 min and the estimated mapping capacity from 63.4 to 58.6 ha per flight.

Future work will focus on expanding the number and diversity of field-validated tree instances, particularly for underrepresented species, adopting stricter individual-tree-level or spatially disjoint evaluation protocols, and validating the complete embedded pipeline under real flight conditions. These developments will support a more rigorous assessment of model generalization and the practical deployment of tree species identification on aerial robotic platforms.

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